

ARNAB BHATTACHARYYA

Gopalpur, West Bengal, India

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SUMMARY

Full-Stack Engineer with a B.Tech in AI & ML and experience building scalable web applications, REST APIs, and AI-powered solutions using TypeScript, Python, Node.js, LangChain, and RAG.

EDUCATION

Brainware University

B.Tech in CSE (Artificial Intelligence & Machine Learning) — CGPA: 9.53

Aug 2021 – July 2025

Barasat, West Bengal

EXPERIENCE

Capsitech IT Solution

July 2025 – June 2026

Assistant System Engineer (Software Development) – L1

Jodhpur, Rajasthan – On-site

- Engineered **10+ full-stack features** using **TypeScript** across **Node.js** and **React/Next.js**, building **REST APIs**, **real-time features (Socket.io)**, and optimizing data workflows with **MongoDB** for improved query performance and reduced latency.
- Developed **SEO-optimized UI components** for **CapsiCore**, improving crawlability via **structured metadata**, routing strategies, and form handling using **Zod** and **Formik**.
- Built and maintained **CRM systems (.NET)** for **Acting Office**, contributing to **UK compliance workflows** with cross-functional collaboration.

Aeka Advisors

April 2025 – June 2025

Web Developer Intern

Kolkata, West Bengal – On-site

- Built a 10+ component **SvelteKit** UI system, reducing duplication and improving dev speed.
- Centralized application state with **Svelte stores**, removing prop drilling across 5+ shared UI components.
- Enhanced UI responsiveness for 6+ screen breakpoints, ensuring smooth layouts on mobile, tablet, and desktop.

Cehpoint E-Learning & Cyber Security Solutions

Oct 2024 – Feb 2025

MERN Stack & AI Integration Intern

Remote

- Contributed to full-stack development using the **MERN stack (MongoDB, Express.js, React.js, Node.js)**.
- Enabled secure authentication and real-time database capabilities via **Firebase**, validated with 100+ users.
- Assisted in embedding AI-driven enhancements to elevate automation and user experience.

PROJECTS

Nera | Next.js, Fastify, PostgreSQL, Minio

- Built a **secure file management system** with encrypted storage using **S3-compatible object storage (Minio)**.
- Implemented **JWT authentication (HTTP-only cookies)** and role-based access control for secure file operations.
- Designed **Fastify REST APIs** with **PostgreSQL** for file upload, search, and nested folder management.

Coditor | Next.js, ASP.NET, Docker

- Developed a **containerized code execution platform** for secure multi-language code execution.
- Used **Docker-based isolation** to safely run user code and capture execution output.
- Implemented **CI/CD pipelines** and automated cleanup workflows for reliable deployments.

Talk-to-PDF | Python, LangChain, ChromaDB, Sentence-Transformers

- Built a **RAG-based CLI application** for querying PDF documents using natural language.
- Implemented document chunking, embeddings, and semantic retrieval using **LangChain** and **ChromaDB**.
- Generated source-aware responses with retrieved context and similarity-based search.

SKILLS

Languages JavaScript, TypeScript, Python, C

Domains Web Development, Object Oriented Programming (OOP), DBMS, Data Structures & Algorithms (DSA)

Technologies React.js, Next.js, SvelteKit, Tailwind CSS, Node.js, Express.js, .NET, REST APIs, JWT Authentication, MongoDB, PostgreSQL, MySQL, Zod, Formik

DevOps & Cloud Git, GitHub, GitHub Actions (CI/CD), Docker, Vercel, Render, S3-compatible storage (Minio)

AI/ML & NLP LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, NLP, Scikit-learn, NLTK, SpaCy, OpenCV, Tensorflow, PyTorch

AWARDS & CERTIFICATIONS

- Top 10, HustleX – IIM Lucknow [Certificate] | Finalist, Ideathon – IIM Bangalore [Certificate]
- Intel® Unnati Industrial Training – Business Contract Validation (May – Jul 2024) [Certificate]